WULFTEC WCRT-200

VFD 7 - SECOND OUTFEED CONVEYOR

Industrial Asset Graph Master Evidence & Replacement Packet

Machine	Wulftec International WCRT-200	**Machine no.**	A6882 / Golden State Vintners #6882
Asset	VFD 7 - Second Outfeed Conveyor	**Location**	WCRT-200 electrical/control cabinet
Electrical source	**Panel H2 - D4 - 1P**	**Feeder circuits**	**14 / 16 / 18**
Removed drive	Siemens MICROMASTER Vector	**Replacement**	Allen-Bradley PowerFlex 4, 22A-B8P0N104, Series A
Primary failure	Recurring F231 - output-current measurement imbalance	**Record date**	September 5, 2026

PURPOSE AND EVIDENCE STANDARD

This packet consolidates available field photographs, schematics, rendered reference pages, catalog numbers, parameter readings, terminal mappings, fault history, isolation guidance, and replacement-drive data for ingestion into the Industrial Asset Graph. User-observed field facts are recorded as such. Manufacturer-defined functions are separated from proposed commissioning assignments. Items that still require a meter, nameplate, PLC trace, or functional test are explicitly marked open.

Critical safety boundary

Qualified personnel only. Apply facility lockout/tagout, verify absence of voltage, wait the drive's required discharge interval, and measure DC-bus voltage before contact. Never megger a motor or cable while it remains connected to a VFD. Do not bypass safety interlocks. Keep personnel and product clear during controlled direction and fault-relay tests.

PACKET CONTENTS

Section	Contents	Evidence status
A	Asset identity, feeder, topology, parts and relationships	Front summary
B	Observed symptoms, F231 isolation logic and service chronology	Field observations + manufacturer definition
C	Siemens parameter highlights and full recorded set	Direct keypad transcription
D	Siemens-to-PowerFlex wiring and commissioning cross-reference	Assigned; meter/function checks flagged
E	Original VFD 7 master record	Preserved unchanged as source appendix
F	Photographic and drawing evidence	Original image files rendered with filenames

A. ASSET GRAPH RECORD

Graph field	Value	Confidence / provenance
facility	J. Lieb Foods - Forest Grove, Oregon	Project context
machine	Wulftec International WCRT-200	Manual cover / prior machine record
machine_identifier	A6882; Golden State Vintners #6882	Manual cover and prior record
asset_id	VFD 7	User-confirmed service designation
service	Second Outfeed Conveyor	Control schematic / field report
upstream_source	Panel H2 - D4 - 1P	User-reported September 5, 2026
source_circuits	14, 16, 18	User-reported three-phase circuit positions
legacy_drive	Siemens MICROMASTER Vector	Parameter record and field record
replacement_drive	Allen-Bradley PowerFlex 4; Cat. 22A-B8P0N104; Series A	Nameplate photograph
failure_mode	F231 output-current measurement imbalance	Fault history P140-P143/P930 and field report

RELATIONSHIPS

From	Relationship	To	Details
Panel H2	**FEEDS**	D4-1P / circuits 14-16-18	Three-phase source record
D4-1P	**SUPPLIES**	VFD 7	Incoming wires 591/593/595
VFD 7	**CONTROLS**	Second Outfeed Conveyor motor	Outgoing wires 592/594/596
SLC 5/03 control system	**COMMANDS / MONITORS**	VFD 7	Via field wiring/interposing logic; exact rack/slot/point remains open
VFD 7 fault relay	**REPORTS STATUS**	Machine control	Wires 188/187; PLC destination remains open

POWER / CONTROL TOPOLOGY

PANEL H2	D4-1P	VFD 7	CONVEYOR MOTOR
Circuits 14 / 16 / 18	591 / 593 / 595	PowerFlex 4 22A-B8P0N104	592 / 594 / 596
Upstream distribution	R/L1 - S/L2 - T/L3	U/T1 - V/T2 - W/T3	Second outfeed conveyor

B. EQUIPMENT AND PART IDENTIFICATION

Component	Manufacturer / catalog	Recorded rating / function	Status
Machine	**Wulftec International WCRT-200**	Stretch-wrapper system; manual dated May 2001	Confirmed by manual cover
Legacy VFD	**Siemens MICROMASTER Vector**	VFD 7, second outfeed conveyor; software P922=2.04	Family confirmed; exact VFD 7 order/serial not visible
Reference Siemens drive	**6SE3213-6CA40**	750 W; 208/240 V input; 3-phase 3.90 A output	Photo marked Drive 6 - comparison only
Replacement VFD	**Allen-Bradley PowerFlex 4 22A-B8P0N104, Series A**	1.5 kW / 2 HP; input 3-phase 200-240 VAC, 48-63 Hz, 9.5 A; output 3-phase 0-230 VAC, 0-240 Hz, 8.0 A continuous, 12.0 A/60 s	Confirmed by replacement nameplate
PLC platform	**Allen-Bradley SLC 5/03**	Machine sequence/control; second output module point 4 observed driving a clicking relay	Platform/observation recorded; exact I/O address open
Motor	**Sterling Electric SH0014PCA - second outfeed conveyor motor**	1 HP; 1750 RPM; 3-phase; 60 Hz; 208-230/460 V; 3.2-2.9/1.45 A; 85.5% efficiency; code K; service factor 1.15.	Confirmed by user's direct plate reading and nameplate photograph; glare-obscured fields remain open

AVAILABLE DRAWING / DOCUMENT IDENTIFIERS

Evidence	Identification	Relevant content
General arrangement	Wulftec drawing, revision dated 2000/11/22	Machine footprint and conveyor layout; conveyor speed revision 30 FPM to 36 FPM
Control schematic	Field photograph IMG_2736(1).jpeg	VFD-4 through VFD-8, including second outfeed conveyor VFD-7 and terminal/wire annotations
PLC input drawing	Field photograph IMG_2733.jpeg	Reverse, stop, safety and oversize-related input circuits
Machine manual	WCRT-200 Instruction Guide and Operating Manual	Golden State Vintners #6882; May 2001; Wulftec contact data printed on cover
Siemens instructions	G85139-H1751-U529-D1	MICROMASTER Vector / MIDIMASTER Vector operating instructions
PowerFlex instructions	Rockwell 22A-UM001	PowerFlex 4 terminal definitions, control examples, programming and fault data

C. FAILURE HISTORY AND DIAGNOSTIC RECORD

Observation	Recorded detail	Interpretation boundary
Recurring fault	Siemens F231; history P140=231, P141=231, P142=231, P143=231 and P930=231	Repeated stored fault, not a one-time display artifact
Reset behavior	P/reset clears temporarily; fault returns	Reset does not establish root cause
Cycle timing	VFD 7 observed faulting near the end of the second-half/outfeed sequence	Sequence timing can trigger the condition but does not prove photoeye causation
Isolation result	**VFD 7 reportedly faulted with motor leads disconnected**	Strongly favors internal drive output/current-sensing hardware if U/V/W were removed at the drive and test conditions were equivalent
Related system behavior	Reset-button operation/chatter and other drives' faults were observed during troubleshooting	May indicate shared control/power effects; do not merge distinct fault events without timestamps
Photoeye correlation	PE10 blockage/requested motion correlated with faults during earlier testing	A command device may expose the fault by commanding run; F231 itself remains an output-current measurement fault

F231 ISOLATION DECISION

Test state	Result	Evidence weight / next action
U/V/W disconnected at VFD; PE retained	**F231 remains**	Strong evidence for inverter output stage/current sensing or internal hardware. Stop repeated run attempts; replace/bench-evaluate drive.
U/V/W disconnected at VFD; PE retained	**F231 clears**	Investigate cable, motor windings, terminations, moisture/ground leakage, mechanical load and motor data.
Test configuration uncertain	Any outcome	Repeat only under controlled, documented conditions before declaring root cause.

DE-ENERGIZED CHECKS TO RETAIN

Phase-to-phase resistance U-V, V-W, W-U; each phase to PE; insulation resistance only with VFD and sensitive electronics isolated; mechanical freedom; bearing/chain condition; terminal torque/discoloration; moisture or contamination; incoming phase balance; commanded control voltage and simultaneous-command check.

Working conclusion

The reported persistence of F231 with the outgoing motor leads disconnected materially supports replacing the Siemens drive. Preserve the removed drive for inspection until the PowerFlex completes direction, current, stop, fault-relay, and repeated-cycle verification.

D. COMPLETE WIRING MIGRATION SUMMARY

Wire	Legacy Siemens landing	PowerFlex 4 landing	Function	Verification
591	A / L1	**R / L1**	Incoming phase 1	Panel H2-D4-1P; circuit set 14/16/18
593	B / L2	**S / L2**	Incoming phase 2	Confirm phase-to-phase voltage
595	C / L3	**T / L3**	Incoming phase 3	Confirm phase sequence
592	U	**U / T1**	Motor phase U	Direction test
594	V	**V / T2**	Motor phase V	Direction test
596	W	**W / T3**	Motor phase W	Direction test
6	T2 - 0 V/common	**04 - Digital Common**	Control reference	PowerFlex selector SRC after meter check
129	T6 - DIN2; P052=18	**02 - Start/Run FWD**	Forward; Siemens FF4=56 Hz	Confirm +24 V to wire 6 when active
128	T5 - DIN1; P051=18	**03 - Direction/Run REV**	Reverse; Siemens FF5=55 Hz	Confirm +24 V to wire 6 when active
188	T19 - RL1 N.O.	**R1 - Relay N.O.**	Fault-status switched leg	Program and prove fail-safe state
187	T20 - RL1 common	**R2 - Relay Common**	Fault-status common	R3 remains unused unless redesigned
Not recorded	PE	**PE / ground**	Protective earth	Identify and verify continuity

SNK / SRC AND DIP-SWITCH RECORD

Item	Recorded setting	Meaning
Siemens 5-switch bank	**1 UP; 2 DOWN; 3 DOWN; 4 UP; 5 DOWN**	Analog-input configuration. Not equivalent to the PowerFlex SNK/SRC selector.
PowerFlex SNK/SRC	**SRC - meter verify**	Cover states SRC inputs activate when pulled to +24 V. Functional match if 128/129 are +24 V relative to wire 6 when active.
PowerFlex factory jumper	01 STOP to 11 +24 V	Shown installed in photographs. Retain/remove only as required by the selected start scheme and safety design.

E. PARAMETER RECORD - CRITICAL VALUES

Parameter	Value	Meaning / migration relevance
P002 / P003	**1.5 s / 1.0 s**	Acceleration / deceleration
P006 / P007	**2 / 0**	Fixed-frequency source; terminal control
P013	**60 Hz**	Maximum frequency
P044 / P046	**56 Hz / 55 Hz**	Fixed frequency 4 forward / fixed frequency 5 reversed by P050
P051 / P052	**18 / 18**	DIN1 and DIN2 each provide run plus fixed-frequency selection
P061 / P062	**6 / 8**	RL1 fault indication / RL2 warning active
P080-P085	**0.80; 60 Hz; 1725 RPM; 3.2 A; 230 V; 1 HP**	Recorded Siemens motor data. Verify against the actual VFD 7 motor nameplate before entering PowerFlex values.
P086 / P186	**150% / 200%**	Motor current limit / instantaneous current limit
P089	**4.6 ohm**	Recorded stator resistance
P111 / P112 / P113	**1.01 HP / 6 / 14**	Inverter rating/type/model code: MICROMASTER Vector MMV75/2
P128	**120 s**	Fan switch-off delay
P134	**303 VDC**	Recorded DC-link snapshot
P140-P143 / P930	**231 / 231 / 231 / 231 / 231**	Repeated F231 fault history
P922	**2.04**	Software version
P944 / P971	**0 / 1**	Factory reset inactive / EEPROM storage enabled

POWERFLEX COMMISSIONING PARAMETERS

Legacy motor-data comparison: Siemens P085=1 HP matches the Sterling motor's 1 HP rating. P083=3.2 A matches the motor's 208 V full-load current; the plate lists 2.9 A at 230 V and 1.45 A at 460 V. P082=1725 RPM differs slightly from the plate's 1750 RPM and should be corrected to 1750 RPM in the replacement drive. Set voltage/current as a matched pair after verifying the actual applied motor voltage.

Parameter	Required action	Reason
P031-P035	Enter 60 Hz and the matched nameplate pair: 208 V / 3.2 A or 230 V / 2.9 A, based on measured system voltage; use 0 Hz minimum and 60 Hz maximum unless machine design requires otherwise	Do not mix voltage and current columns or use the drive's 8.0 A rating as motor FLA
P036	Select actual maintained two-wire forward/reverse start source	Terminals 02 and 03 will command direction
P037	Match stop mode to machine design	Terminal 01 behavior depends on start-source selection
P038	Establish replacement speed reference	PowerFlex direction inputs do not inherently reproduce separate 56/55 Hz speeds
P039 / P040	Set 1.5 s accel / 1.0 s decel after machine review	Matches recorded Siemens ramps
A051 / A052	Assign only if separate preset-speed/interlock functions are deliberately required	Avoid unintended override behavior
A055	Set relay output for Fault and prove actual logic	Preserve machine fault-status behavior on R1/R2

F. COMMISSIONING / ACCEPTANCE RECORD

Check	Required result / record	Status
Identity	PowerFlex catalog photographed; motor identified as Sterling Electric SH0014PCA. Capture a glare-free straight-on image for remaining obscured plate fields.	**PARTIAL**
Feeder	Verify Panel H2 - D4 - 1P and circuits 14/16/18 against panel directory and disconnect	**OPEN**
Power	Record L1-L2, L2-L3, L3-L1; verify grounding and phase rotation	**OPEN**
Control mode	Wire 6 = 0 V/common; active 128/129 approximately +24 V to wire 6; selector at SRC	**OPEN**
Command exclusivity	128 and 129 never energized simultaneously	**OPEN**
Forward	129/terminal 02 produces correct forward motion	**PASS / FAIL**
Reverse	128/terminal 03 produces correct reverse motion	**PASS / FAIL**
Stop / safety	Terminal 01, normal stop, E-stop and all guards/interlocks operate as designed	**PASS / FAIL**
Fault relay	R1/R2 provides correct healthy/fault state to machine control	**PASS / FAIL**
Motor current	Record steady current per phase; compare with 3.2 A at 208 V or 2.9 A at 230 V	**_____ A**
Cycle test	Run repeated full cycles including end-of-cycle outfeed transition	**_____ cycles**
PLC trace	Record rack, slot, point/address and relay ID for wires 128, 129, 187 and 188	**OPEN**

Technician		**Date/time**	
Reviewer		**Production witness**	
Final disposition	Accepted / correction required / removed from service	**Work order**	

OPEN INFORMATION REQUIRED FOR COMPLETE GRAPH CLOSURE

Installed PowerFlex serial number; glare-free image of remaining obscured motor nameplate values; enclosure/cabinet tag; disconnect/fuse/breaker catalog and ratings; verified PE wire ID; exact SLC rack/slot/point and logical address; interposing relay IDs; drawing rung/page; final programmed PowerFlex parameter export; measured voltages/currents; final root cause; removed-drive disposition; post-replacement cycle count and sign-off.

G. SOURCE REGISTER AND INGESTION NOTES

Source	Included as	Use in Industrial Asset Graph
Siemens VFD 7 Second Outfeed Conveyor Master Record	Appendix 1 - original 10-page PDF	Failure analysis, safety, legacy identification, manual extracts and field worksheets
Siemens-to-Allen-Bradley Completed Cross-Reference	Appendix 2 - original 6-page PDF	Full parameter record plus replacement terminal mapping and setup record
Machine drawings and manual cover	Appendix 3 - original photographs	Machine/document nodes with image evidence and drawing relationships
PowerFlex field photographs	Appendix 3 - original photographs	Part identity, terminal map, fault list, control cover and physical condition

RECOMMENDED GRAPH NODES

Machine: WCRT-200 A6882. Area/location: WCRT-200 cabinet and second outfeed conveyor. Distribution: Panel H2, feeder D4-1P, circuits 14/16/18. Drive asset: VFD 7. Legacy part instance: Siemens MICROMASTER Vector. Replacement part instance: PowerFlex 4 22A-B8P0N104. Motor asset: second outfeed conveyor motor. Controller: Allen-Bradley SLC 5/03. Signals: forward, reverse, digital common and drive fault. Documents: Siemens manual, Rockwell manual, Wulftec operating manual, general arrangement, PLC input sheet, conveyor-control sheet, parameter record and this packet. Event: recurring F231 and drive replacement.

RECOMMENDED RELATIONSHIP TYPES

FEEDS, PROTECTS, LOCATED_IN, CONTROLS, DRIVES, COMMANDS, MONITORS, REPORTS_FAULT_TO, DOCUMENTED_BY, REPLACED_BY, HAS_PARAMETER_SET, EXPERIENCED_FAULT, TESTED_BY, VERIFIED_BY and REQUIRES_FOLLOWUP.

EVIDENCE LABELS

Use CONFIRMED for direct nameplate, terminal-label, user field confirmation or direct parameter capture. Use REFERENCE_ONLY for photographs explicitly marked as adjacent Drive 6. Use PROPOSED for the PowerFlex terminal/programming migration until meter and functional tests pass. Use OPEN for missing PLC addresses, glare-obscured motor fields, protective-device data and final commissioning measurements.

Appendix ordering

Appendix 1 preserves the earlier Siemens VFD 7 master record. Appendix 2 preserves the completed replacement cross-reference and the full recorded parameter list. Appendix 3 renders every currently available relevant original photograph with its source filename. Duplicate views are retained when they are separate source files.

SIEMENS VFD 7

SECOND OUTFEED CONVEYOR

Asset field	Current record
Asset ID	VFD 7
Service	Second outfeed conveyor
Manufacturer / family	Siemens MICROMASTER Vector
Recurring reported fault	F231 - output-current measurement imbalance
Document status	Working master record; field verification items remain open
Prepared	September 5, 2026

SAFETY

Lock out and tag out all applicable energy sources before moving power or motor conductors. Verify absence of voltage with an adequately rated tester. The drive label requires at least five minutes after power removal for internal discharge; verify the DC bus is discharged before contact. Do not bypass guards or safety interlocks.

Evidence boundary

Confirmed for VFD 7: asset number, second-outfeed-conveyor service, and recurring F231 report. **Not yet confirmed from a VFD 7 nameplate or terminal photograph:** exact catalog number, serial number, horsepower, and wire numbers. The supplied nameplate is visibly marked **6**; its specifications and wiring are retained in this packet only as an adjacent/reference-drive comparison.

1. VFD 7 Field Identification

Item	Enter/verify at machine
Manufacturer	Siemens
Product family	MICROMASTER Vector
Catalog/order number	______________________________
Serial number	______________________________
Motor rating	________ HP / ________ kW
Input rating	________ V, ______ phase, ______ A
Output rating	________ V, 3 phase, ______ A
Location/cabinet	______________________________
Driven motor tag	______________________________

Reported operating symptom

F231 occurs on VFD 7, serving the second outfeed conveyor. F231 is defined by Siemens as **output-current measurement imbalance**. A reset can clear the indication temporarily, but a recurring fault requires isolation of the motor/cable side from the inverter itself.

Do not assign a photoeye as the cause solely because a photoeye initiates conveyor motion. A command device can expose the fault by requesting a run, but F231 specifically concerns the drive output-current measurement path.

Reference documentation

Siemens MICROMASTER Vector / MIDIMASTER Vector Operating Instructions, document family G85139-H1751-U529. The Siemens fault table directs F231 troubleshooting to the F002 checks: motor and cable shorts/earth faults, motor-data settings, stator resistance, acceleration, boost, and mechanical obstruction/overload.

2. Isolation Test for Recurring F231

Step	Action	Meaning
1	Record exactly when the fault occurs: power-up, run command, acceleration, steady run, or stop.	Separates static/internal faults from load-dependent faults.
2	LOTO. Wait at least five minutes and verify the DC bus is discharged. Photograph and label U, V, W before removal.	Prevents shock and preserves phase order.
3	Disconnect only the motor conductors from VFD 7 U/V/W; individually insulate and secure them. Keep PE connected.	Isolates the motor and outgoing cable from the drive.
4	Re-energize under controlled conditions and issue the same run command. Keep personnel clear; the inverter output is energized even without a motor	Recreates the trigger with the load side removed.
5A	If F231 remains with U/V/W disconnected at the drive, stop testing.	Strongly points to the inverter output/current-sensing hardware or internal c
5B	If F231 disappears, LOTO again and test the outgoing cable and motor.	Points toward cable, terminations, motor windings, or ground leakage.

De-energized motor/cable checks

Check	Record
U-V / V-W / W-U resistance	________ / ________ / ________ ohms; should be closely balanced
Each phase to PE with ordinary ohmmeter	____
Insulation-resistance test	Test only after isolating the cable/motor from the VFD and other electronics. Record test voltage and result
Mechanical condition	Conveyor free/bound: __________ Motor/bearings: __________
Terminations	Loose, discolored, damaged, wet, contaminated: __________________________

Never megger through a connected VFD. Disconnect sensitive electronics before insulation-resistance testing.

3. VFD 7 Wiring Record

Complete this directly from VFD 7. Do not copy the Drive 6 reference numbers unless each terminal is physically verified.

Circuit	Terminal	VFD 7 wire number	Destination / notes
Control common	2	____________	0 V/common
Digital input 1	5	____________	Function verified by P051: __________
Digital input 2	6	____________	Function verified by P052: __________
Relay 1	18	____________	P061 function: __________
Relay 1	19	____________	Contact role: __________
Relay 1	20	____________	Contact role: __________
Incoming phase A / L1	A/L1	____________	Voltage to B/C: __________
Incoming phase B / L2	B/L2	____________	Voltage to A/C: __________
Incoming phase C / L3	C/L3	____________	Voltage to A/B: __________
Motor phase	U	____________	
Motor phase	V	____________	
Motor phase	W	____________	
Protective earth	PE	____________	Continuity verified: ______

Adjacent/reference Drive 6 - verbally confirmed

Terminal/function	Wire
2 / 0 V common	6
5 / digital input 1	128
6 / digital input 2	129
19 / fault-relay contact	188
20 / fault-relay contact	187
Incoming A-B-C	591 - 593 - 595
Outgoing U-V-W	592 - 594 - 596

4. Parameters to Capture Before Any Change

Parameter	Function	VFD 7	Good comparison drive
P001	Display selection	______	______
P002	Ramp-up time	______ s	______ s
P003	Ramp-down time	______ s	______ s
P005	Digital frequency setpoint	______ Hz	______ Hz
P006	Frequency-command source	______	______
P007	Keypad control	______	______
P009	Parameter protection	______	______
P012 / P013	Min / max frequency	____ / ____ Hz	____ / ____ Hz
P051 / P052	Digital-input functions	____ / ____	____ / ____
P061 / P062	Relay-output functions	____ / ____	____ / ____
P080	Motor power factor	______	______
P081	Motor nameplate frequency	______ Hz	______ Hz
P082	Motor nameplate speed	______ rpm	______ rpm
P083	Motor nameplate current	______ A	______ A
P084	Motor nameplate voltage	______ V	______ V
P085	Motor nameplate power	______ kW	______ kW
P086	Motor current limit	______ %	______ %
P088	Automatic calibration	______	______
P140-P143	Fault history	________________	________________

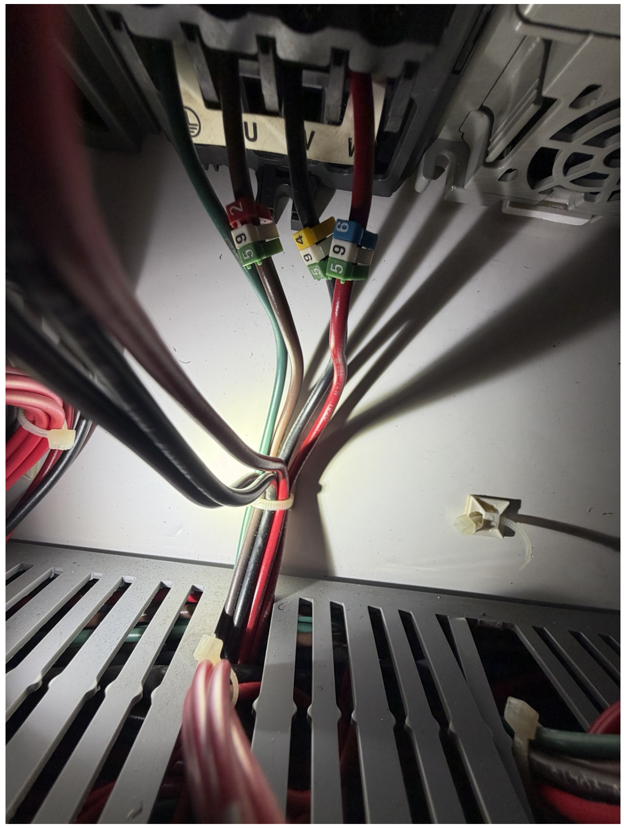
Parameter navigation

Press P to enter parameters, use the arrow buttons to select a parameter, and press P to display its value. Pressing P after editing stores the new value. For documentation, record values without changing them.

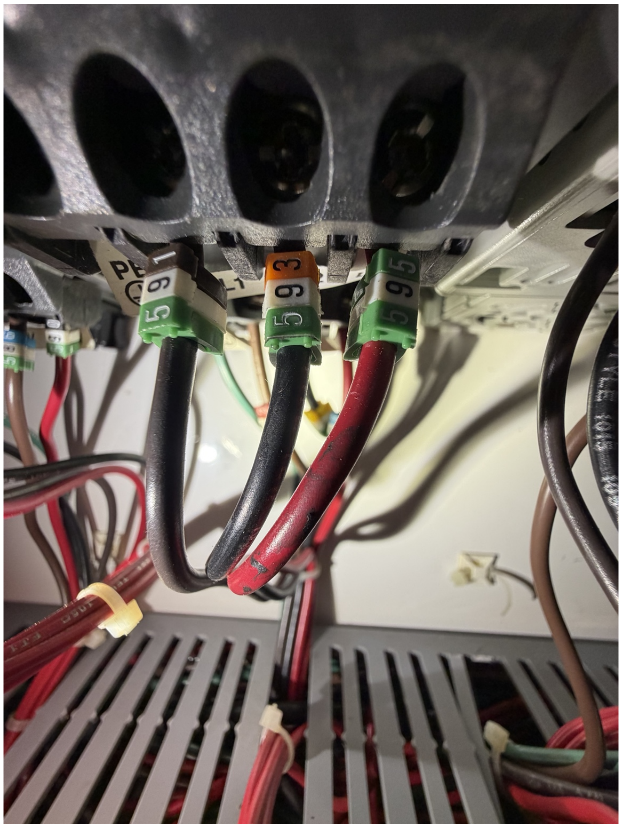
5. Photographic Evidence - Drive 6 Reference

These photographs document the adjacent/reference drive currently marked 6. They illustrate terminal locations and provide comparison information; they do not independently establish VFD 7 wiring.

Drive marked 6. Nameplate: 6SE3213-6CA40; 750 W; 208/240 V input; 3-phase 3.90 A output.



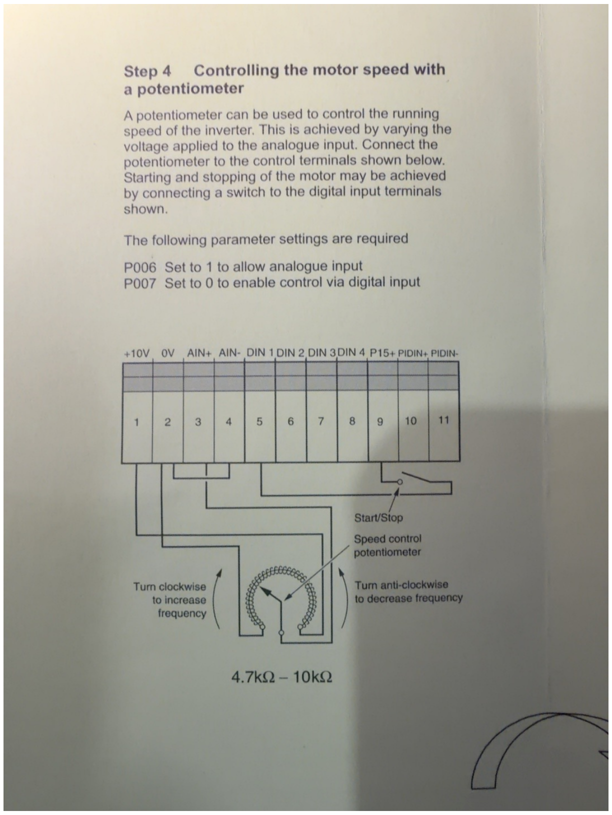
Reference U/V/W motor-output conductors, verbally confirmed as 592/594/596.



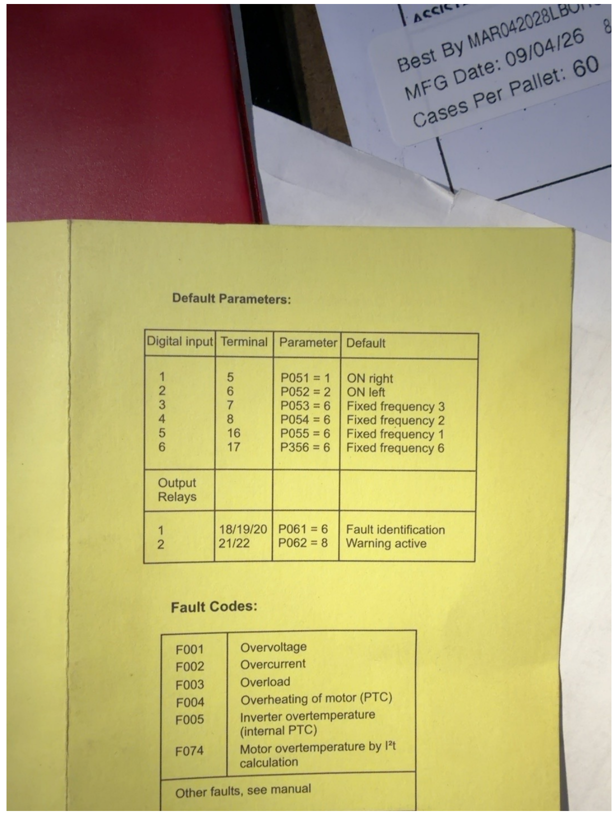
Reference incoming phase conductors, verbally confirmed as 591/593/595.

Reference control terminal block showing wires 6, 128, 129, 188 and 187.

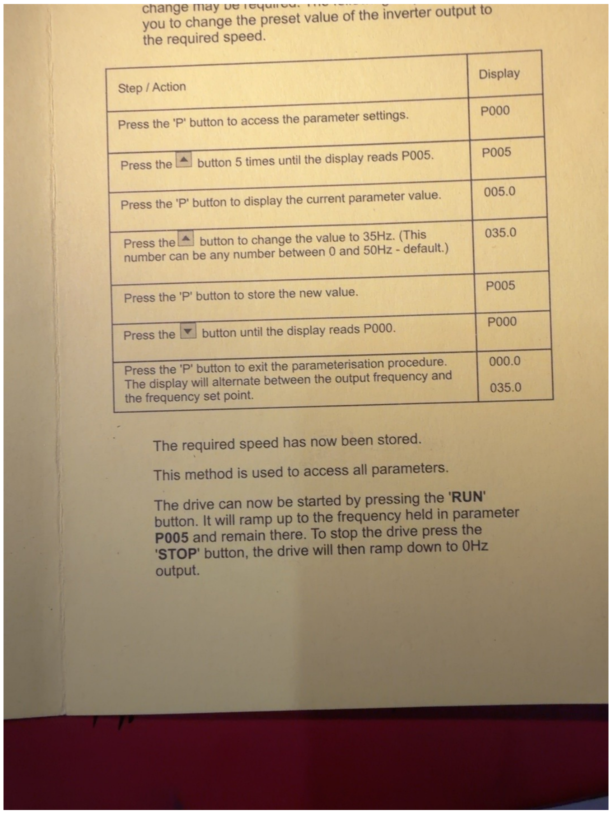
6. Manual Evidence and Close-ups



Manual terminal map: 2 = 0 V, 5 = DIN1, 6 = DIN2.



Manual defaults: terminals 18/19/20 are relay 1; P061 default 6 = fault indication.



change may be required. ...
you to change the preset value of the inverter output to the required speed.

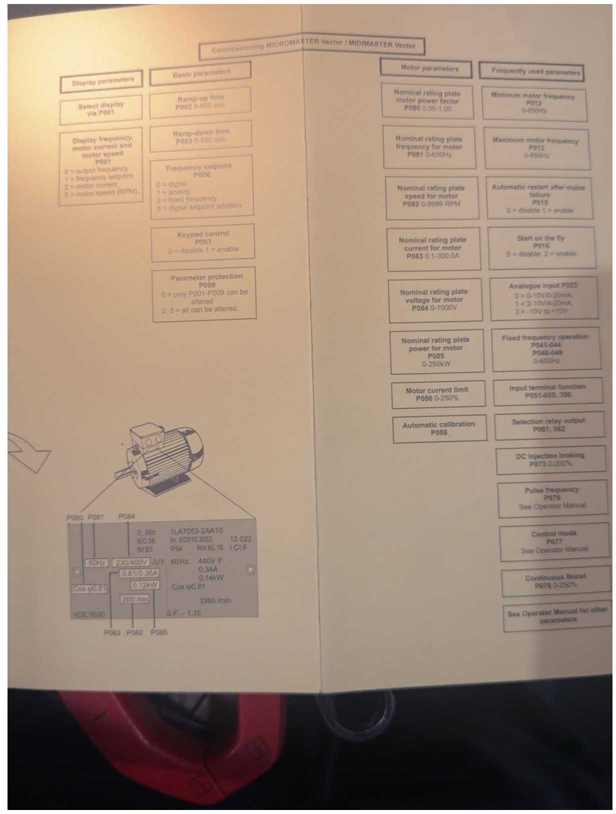
Step / Action	Display
Press the 'P' button to access the parameter settings.	P000
Press the ▲ button 5 times until the display reads P005.	P005
Press the 'P' button to display the current parameter value.	005.0
Press the ▲ button to change the value to 35Hz. (This number can be any number between 0 and 50Hz - default.)	035.0
Press the 'P' button to store the new value.	P005
Press the ▼ button until the display reads P000.	P000
Press the 'P' button to exit the parameterisation procedure. The display will alternate between the output frequency and the frequency set point.	000.0 035.0

The required speed has now been stored.

This method is used to access all parameters.

The drive can now be started by pressing the **'RUN'** button. It will ramp up to the frequency held in parameter **P005** and remain there. To stop the drive press the **'STOP'** button, the drive will then ramp down to 0Hz output.

Manual parameter-navigation example.



Manual commissioning parameter overview.

7. Service Log and Final Disposition

Date/time	Operating event	Fault	Test/result	Technician
____________	________________	_______	________________________	__________
____________	________________	_______	________________________	__________
____________	________________	_______	________________________	__________
____________	________________	_______	________________________	__________

Disposition checklist

Item	Status / result
VFD 7 nameplate photographed	____
VFD 7 wiring physically verified	____
Fault reproduced with exact event recorded	____
U/V/W disconnected-at-drive isolation test	Fault remained / cleared / not performed
Motor/cable resistance and insulation results	____
Parameters compared to matched drive	____
Root cause	____
Corrective action	____
Production verification: cycles / duration	____

Working conclusion

If VFD 7 repeats F231 with its motor leads physically removed at U/V/W, the evidence strongly favors an internal inverter/current-measurement failure. If the fault clears, continue on the outgoing motor cable, motor windings, terminations, leakage to ground, and mechanical load. Replace or condemn equipment only after recording the isolation result and verifying the exact VFD 7 identity.

Technician: _________________________________ Date: _______________ Reviewer: _________________________________

SIEMENS MICROMASTER VECTOR

Complete Recorded Parameter Set - VFD 7 / Second Outfeed Conveyor

Field values transcribed from the operator's direct keypad readings. Values are recorded settings or live read-only snapshots, not assumed factory defaults.

P Number	Value	Siemens function / interpretation
P001	**0**	Display mode - output frequency
P002	**1.5**	Ramp-up time (s)
P003	**1**	Ramp-down time (s)
P004	**0**	Smoothing time (s)
P005	**5**	Digital frequency setpoint (Hz)
P006	**2**	Frequency source - fixed frequencies
P007	**0**	Keypad control disabled / terminal control
P009	**3**	All parameters readable and adjustable
P010	**1**	Display scaling
P011	**0**	Frequency setpoint memory disabled
P012	**0**	Minimum motor frequency (Hz)
P013	**60**	Maximum motor frequency (Hz)
P014	**0**	Skip frequency 1 (Hz)
P015	**0**	Automatic restart after mains failure disabled
P016	**0**	Start on the fly disabled
P017	**1**	Continuous smoothing
P018	**0**	Automatic restart after fault disabled
P019	**2**	Skip-frequency bandwidth (Hz)
P021	**0**	Minimum analogue frequency (Hz)
P022	**60**	Maximum analogue frequency (Hz)
P023	**0**	Analogue input 1 type - 0-10 V / 0-20 mA
P024	**0**	No analogue setpoint addition
P025	**0**	Analogue output 1 - output frequency
P027	**0**	Skip frequency 2 (Hz)
P028	**0**	Skip frequency 3 (Hz)
P029	**0**	Skip frequency 4 (Hz)
P031	**60**	Jog frequency right (Hz)
P032	**5**	Jog frequency left (Hz)
P033	**1**	Jog ramp-up time (s)
P034	**1**	Jog ramp-down time (s)
P040	**0**	Positioning function disabled
P041	**0**	Fixed frequency 1 (Hz)
P042	**0**	Fixed frequency 2 (Hz)
P043	**0**	Fixed frequency 3 (Hz)
P044	**56**	Fixed frequency 4 (Hz)
P045	**0**	Fixed-frequency 1-4 inversion selection
P046	**55**	Fixed frequency 5 (Hz)

P Number	Value	Siemens function / interpretation
P047	0	Fixed frequency 6 (Hz)
P048	0	Fixed frequency 7 (Hz)
P049	0	Fixed frequency 8 (Hz)
P050	1	Fixed-frequency 5 inverted; 6-8 normal
P051	18	DIN1 terminal 5 - fixed frequency 5 plus RUN
P052	18	DIN2 terminal 6 - fixed frequency 4 plus RUN
P053	0	DIN3 terminal 7 disabled
P054	0	DIN4 terminal 8 disabled
P055	0	DIN5 terminal 16 disabled
P056	0	Digital-input debounce - 12.5 ms
P057	1	Digital-input watchdog interval (s)
P061	6	Relay RL1 - fault indication
P062	8	Relay RL2 - warning active
P063	1	External-brake release delay (s)
P064	1	External-brake stopping time (s)
P065	1	Current threshold for relay (A)
P066	100	Compound braking (%)
P069	1	Ramp extension enabled
P070	4	Braking-resistor duty cycle - 100% (MMV only)
P071	0	Slip compensation (%)
P072	250	Slip limit (%)
P073	0	DC injection braking (%)
P074	1	I2t motor protection curve
P075	0	External braking resistor not connected
P076	0	Pulse frequency - 16 kHz, normal modulation
P077	1	FCC control mode
P078	100	Continuous boost (%)
P079	50	Starting boost (%)
P080	0.80	Motor power factor
P081	60	Motor nameplate frequency (Hz)
P082	1725	Motor nameplate speed (RPM)
P083	3.2	Motor nameplate current (A)
P084	230	Motor nameplate voltage (V)
P085	1	Motor nameplate power (hp with P101=1)
P086	150	Motor current limit (%)
P087	0	Motor PTC disabled
P088	0	Automatic stator calibration inactive
P089	4.6	Stator resistance (ohm)
P091	0	USS serial slave address
P092	6	USS baud rate - 9600
P093	0	Serial timeout disabled
P094	60	USS nominal system setpoint (Hz)

P Number	Value	Siemens function / interpretation
P095	**0**	USS compatibility - 0.1 Hz resolution
P099	**0**	No option module
P101	**1**	North American operation - 60 Hz / hp
P111	**1.01**	Inverter power rating (hp), read-only
P112	**6**	Inverter type - MICROMASTER Vector
P113	**14**	Drive model - MMV75/2
P121	**1**	RUN button enabled when P007=1
P122	**1**	Forward/reverse button enabled when P007=1
P123	**1**	JOG button enabled when P007=1
P124	**1**	Up/down buttons enabled when P007=1
P125	**1**	Forward and reverse operation allowed
P128	**120**	Fan switch-off delay (s)
P131	**0**	Frequency setpoint (Hz), read-only snapshot
P132	**0**	Motor current (A), read-only snapshot
P133	**0**	Motor torque (%), read-only snapshot
P134	**303**	DC-link voltage (V), read-only snapshot
P135	**0**	Motor speed (RPM), read-only snapshot
P137	**0**	Output voltage (V), read-only snapshot
P138	**0**	Rotor/shaft frequency (Hz), read-only snapshot
P139	**0**	Peak output-current record
P140	**231**	Most recent fault - F231
P141	**231**	Previous fault - F231
P142	**231**	Second previous fault - F231
P143	**231**	Third previous fault - F231
P186	**200**	Instantaneous motor-current limit (%)
P201	**0**	PID closed-loop mode disabled
P202	**1**	PID proportional gain
P203	**0**	PID integral gain
P204	**0**	PID derivative gain
P205	**1**	PID sample interval - 25 ms
P206	**0**	PID transducer filtering off
P207	**100**	PID integral capture range (%)
P208	**0**	Transducer type - direct acting
P210	**1.4**	Transducer reading (%), read-only snapshot
P211	**0**	PID 0% setpoint
P212	**100**	PID 100% setpoint
P220	**0**	Frequency cutoff disabled
P321	**0**	Analogue-input-2 minimum frequency (Hz)
P322	**60**	Analogue-input-2 maximum frequency (Hz)
P323	**0**	Analogue input 2 - 0-10 V / 0-20 mA
P356	**6**	DIN6 terminal 17 - fixed frequency 6
P386	**1**	Vector speed-loop proportional gain

P Number	Value	Siemens function / interpretation
P387	**1**	Vector speed-loop integral gain
P720	**0**	Direct I/O functions - normal operation
P721	**0**	Analogue-input-1 voltage (V), read-only
P722	**0**	Analogue-output-1 current (mA)
P723	**0**	Digital-input state (hex) - all inputs off
P724	**0**	Direct relay control - both off
P725	**0**	Analogue-input-2 voltage (V), read-only
P910	**0**	Local control mode
P922	**2.04**	Software version, read-only
P923	**0**	Equipment system/reference number
P930	**231**	Most recent fault - F231 (copy of P140)
P931	**0**	Most recent warning - none
P944	**0**	Factory reset inactive
P971	**1**	EEPROM storage enabled

Critical retained readings: P082 = 1725 RPM; P128 = 120 s; P922 = software 2.04; P140-P143 and P930 = F231. P944 must remain 0 unless an intentional factory reset is required.

Literature: Siemens MICROMASTER Vector / MIDIMASTER Vector Operating Instructions, G85139-H1751-U529-D1; compact guide G85139-H1751-U568-D1, May 1999.

VFD 7 - SIEMENS TO ALLEN-BRADLEY

Completed replacement-drive wiring cross-reference | Second Outfeed Conveyor

Replacement drive	Allen-Bradley PowerFlex 4	**Catalog**	22A-B8P0N104, Series A
Motor rating	1.5 kW / 2.0 HP	**Input**	3-phase, 200-240 V AC, 48-63 Hz, 9.5 A
Output	3-phase, 0-230 V AC, 0-240 Hz	**Output current**	8.0 A continuous; 12.0 A / 60 s

CONTROL-CONDUCTOR CROSS-REFERENCE

Wire	Siemens MICROMASTER Vector	PowerFlex 4 landing	Function / commissioning note	Status
6	T2 - control 0 V	**04 - Digital Common**	Digital-input common. Confirm the SNK/SRC selector and control-voltage scheme before energizing.	**LANDING ASSIGNED VERIFY POLARITY**
129	T6 - DIN2 P052=18; 56 Hz forward	**02 - Start / Run FWD**	Two-wire forward run command. PowerFlex P036 must be configured for the actual two-wire scheme.	**LANDING ASSIGNED PROGRAM + TEST**
128	T5 - DIN1 P051=18; 55 Hz reverse	**03 - Direction / Run REV**	Two-wire reverse run command. Do not permit terminals 02 and 03 to be active together.	**LANDING ASSIGNED PROGRAM + TEST**
188	T19 - RL1 N.O. P061=6 fault	**R1 - Relay N.O.**	Fault-status switched leg. Set PowerFlex relay output selection to Fault and prove the desired energized/de-energized behavior.	**LANDING ASSIGNED PROVE LOGIC**
187	T20 - RL1 common	**R2 - Relay Common**	Fault-status relay common. R3 is the unused normally-closed contact unless the control circuit is intentionally redesigned.	**LANDING ASSIGNED PROVE LOGIC**

POWER AND MOTOR CONDUCTORS

Wire	Siemens	PowerFlex 4	Circuit	Status
591	A / L1	**R / L1**	Incoming line phase 1	**DIRECT CROSS-REFERENCE**
593	B / L2	**S / L2**	Incoming line phase 2	**DIRECT CROSS-REFERENCE**
595	C / L3	**T / L3**	Incoming line phase 3	**DIRECT CROSS-REFERENCE**
592	U	**U / T1**	Motor output phase U	**DIRECT CROSS-REFERENCE**
594	V	**V / T2**	Motor output phase V	**DIRECT CROSS-REFERENCE**
596	W	**W / T3**	Motor output phase W	**DIRECT CROSS-REFERENCE**
Not recorded	PE	**PE / ground**	Equipment grounding conductor	**IDENTIFY IN FIELD**

Important: This is a terminal/function cross-reference, not authorization to energize. Confirm every wire marker and terminal de-energized under LOTO. Verify phase sequence, motor direction, insulation to ground with the VFD disconnected, control polarity, and relay fail-safe behavior before commissioning.

POWERFLEX 4 TERMINAL & SETUP RECORD

Use during installation and commissioning | Source: drive nameplate/photo and Rockwell 22A-UM001

Observed Siemens DIP bank	**1 UP | 2 DOWN | 3 DOWN | 4 UP | 5 DOWN**
PowerFlex 4 SNK/SRC selector	**SET SRC - METER VERIFY BEFORE LANDING**
Relationship	The Siemens five-switch bank configures analog-input modes and is not a position-for-position replacement. PowerFlex cover: SNK inputs activate when pulled to DIG COM; SRC inputs activate when pulled to +24 V. Wire 6 is the recorded 0 V/common, so SRC is the functional match if wires 128/129 measure approximately +24 V to wire 6 when commanded.

TERMINAL IDENTIFICATION

Terminal	PowerFlex 4 function	Assigned VFD 7 conductor / note
01	Stop / enable input	Factory jumper to terminal 11 is installed in the supplied-drive photo. Retain or remove only as required by the selected start scheme and safety design.
02	Start / Run Forward	**Wire 129 - proposed two-wire forward command**
03	Direction / Run Reverse	**Wire 128 - proposed two-wire reverse command**
04	Digital Common	**Wire 6 - control 0 V/common; set SRC after meter verification**
05	Digital Input 1	Unassigned
06	Digital Input 2	Unassigned
R1	Relay Normally Open	**Wire 188 - fault contact switched leg**
R2	Relay Common	**Wire 187 - fault contact common**
R3	Relay Normally Closed	Unassigned
11	+24 V DC	Factory jumper to terminal 01 shown in photo
12	+10 V DC	Unassigned
13	0-10 V input	Unassigned
14	Analog Common	Unassigned
15	4-20 mA input	Unassigned
16	RS-485 shield	Unassigned

COMMISSIONING ITEMS - MUST BE VERIFIED

- P036 Start Source: configure for the actual maintained two-wire forward/reverse control scheme.
- P037 Stop Mode: match machine stopping requirements; terminal 01 behavior changes with start-source selection.
- P038 Speed Reference and commanded frequency: Siemens used 55 Hz reverse and 56 Hz forward; PowerFlex 4 terminals 02/03 select direction, not two independent speeds. Establish the intended replacement speed strategy before running.
- A051/A052 Digital Input selections: leave unused unless a separate preset-speed or interlock function is deliberately added.
- A055 Relay Out Select: configure for Fault, then prove whether the PLC/control circuit requires closed-on-healthy or closed-on-fault logic.
- Motor nameplate values: enter the actual motor volts, hertz and full-load amps; do not copy the replacement drive's 8.0 A rating as motor FLA.

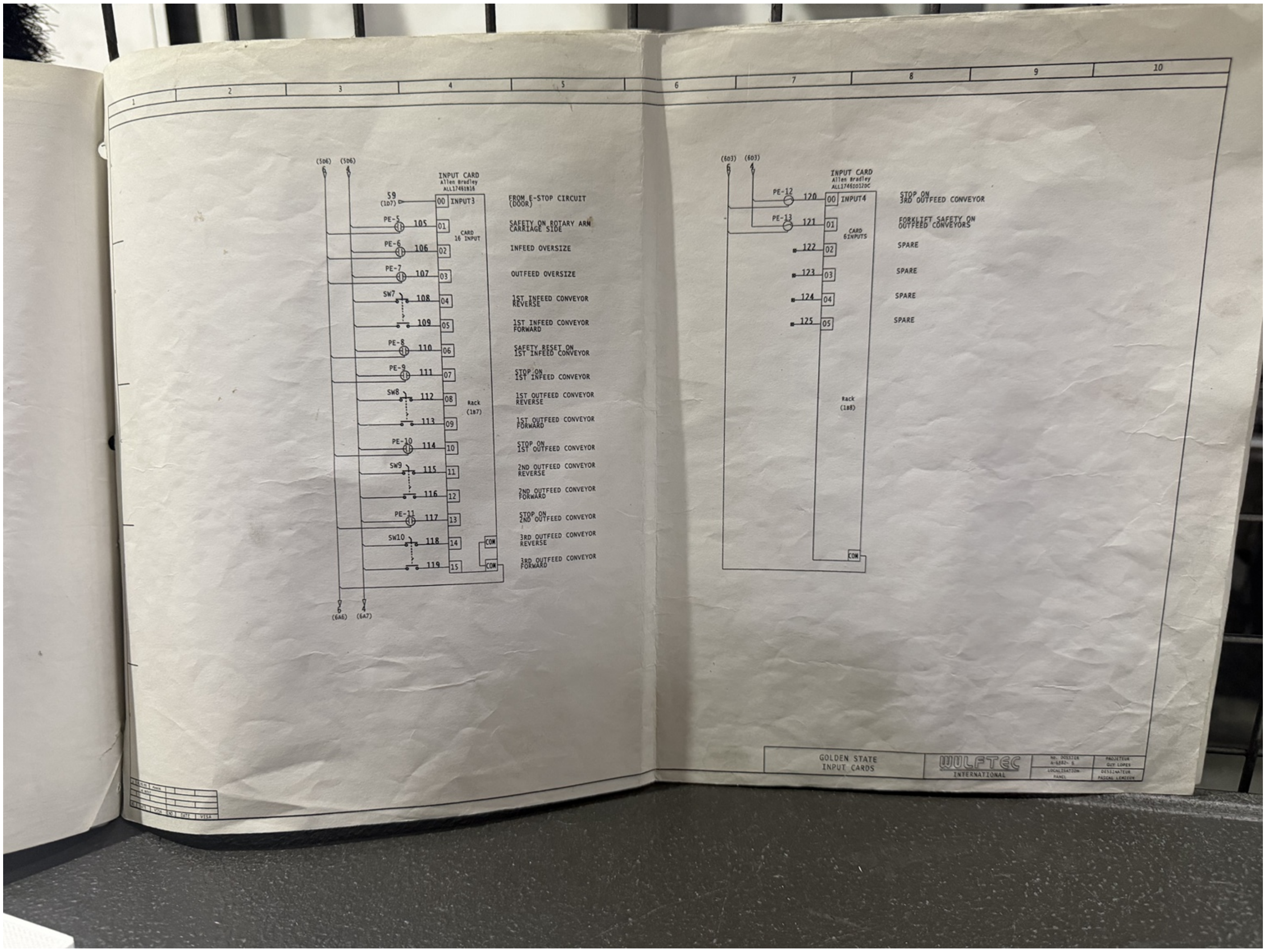
Field verification

Verified by		**Date**	
Control voltage		**SNK / SRC**	
Forward direction	PASS / FAIL	**Reverse direction**	PASS / FAIL
Fault relay logic	PASS / FAIL	**E-stop / stop test**	PASS / FAIL

Do not megger the motor cable while connected to either VFD. Wait the manufacturer's required discharge time, measure DC-bus voltage, and follow the facility electrical-safety procedure before touching terminals.

IMG_2733.jpeg

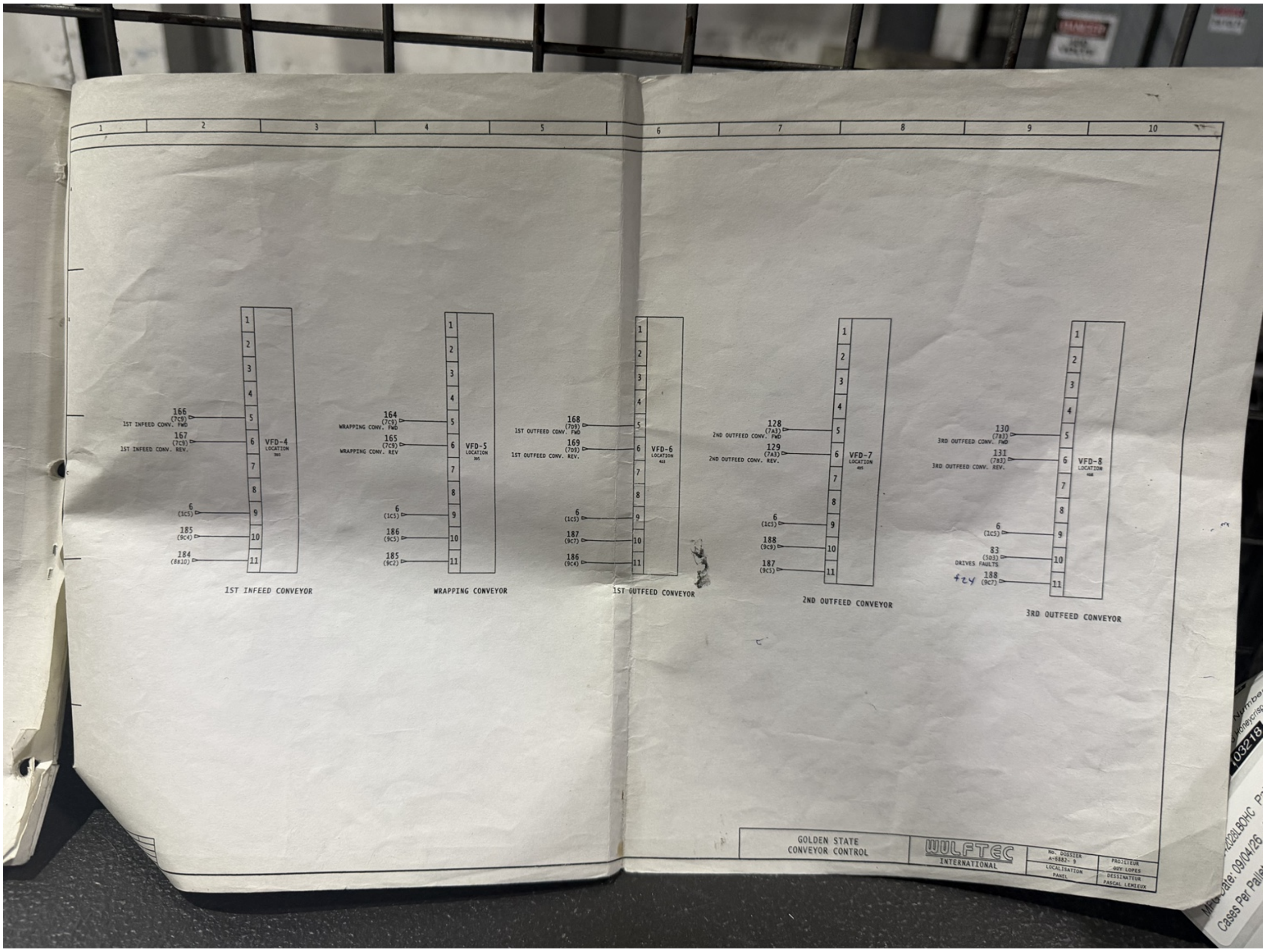
PLC input-card schematic: reverse/stop/safety/oversize circuits.



Source file: IMG_2733.jpeg | SHA-256: d86e2b737e56d8ac16fc405809d2bf1d749535e49a095232580b981fc2ec6865

IMG_2736(1).jpeg

Conveyor-control schematic: VFD-4 through VFD-8, including VFD-7 second outfeed conveyor.



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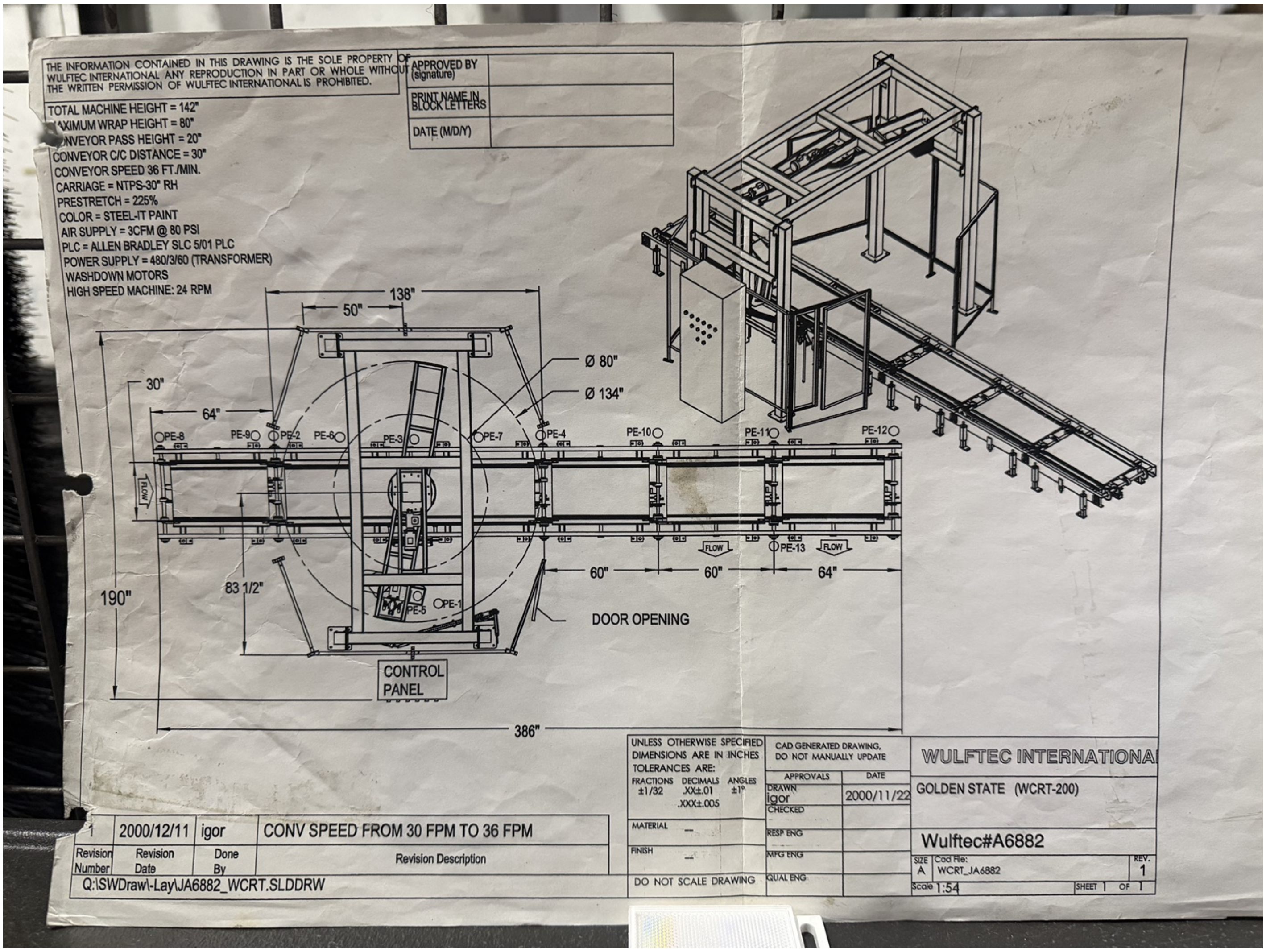
IMG_2740.jpeg

WCRT-200 general arrangement; revision 2000/11/22; conveyor speed changed 30 to 36 FPM.



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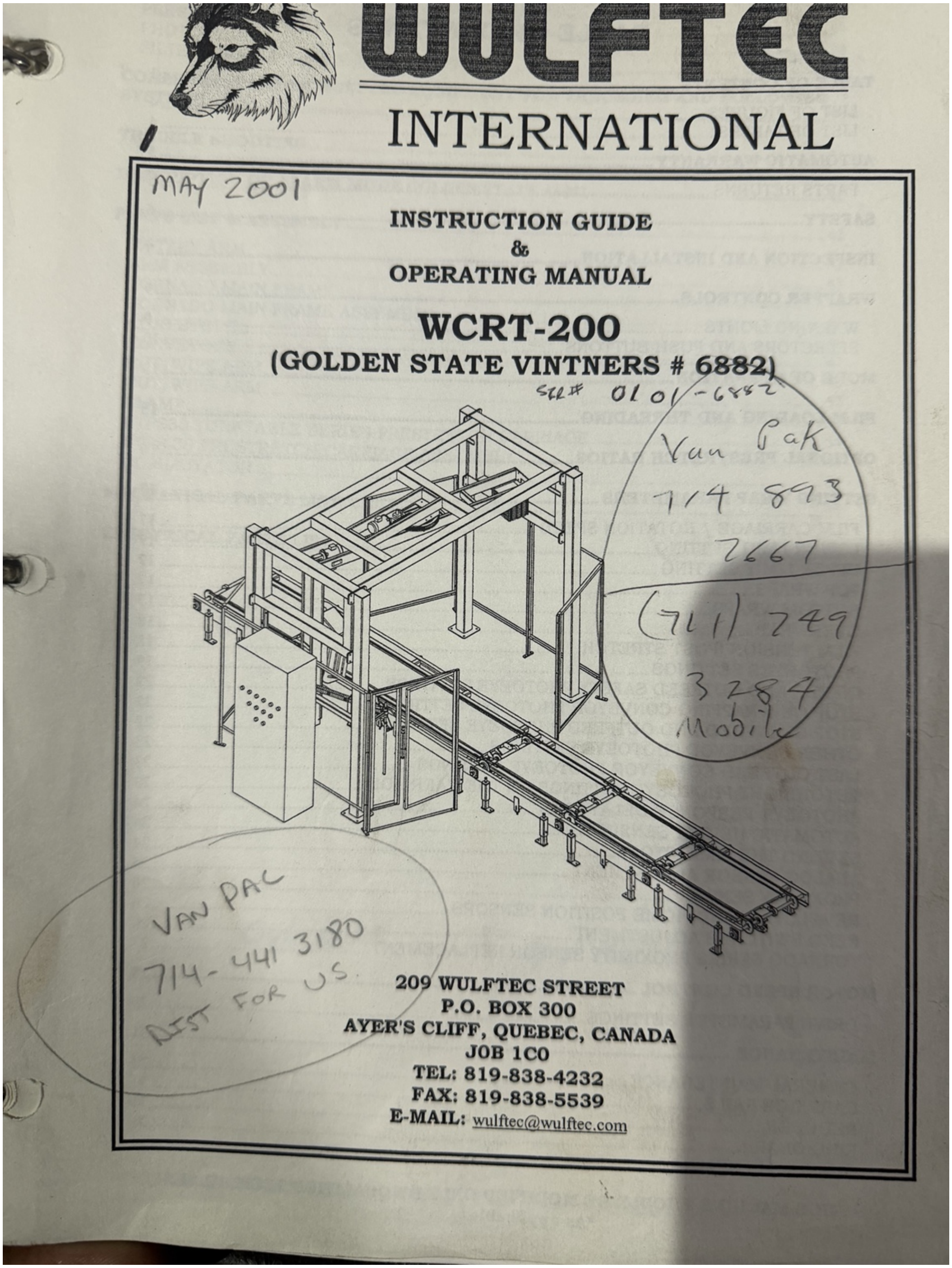
IMG_2747.jpeg

Wulftec WCRT-200 operating manual cover; Golden State Vintners #6882; May 2001.



WULFTEC
INTERNATIONAL

MAY 2001

INSTRUCTION GUIDE
&
OPERATING MANUAL

WCRT-200
(GOLDEN STATE VINTNERS # 6882)

SER# 0101-6882

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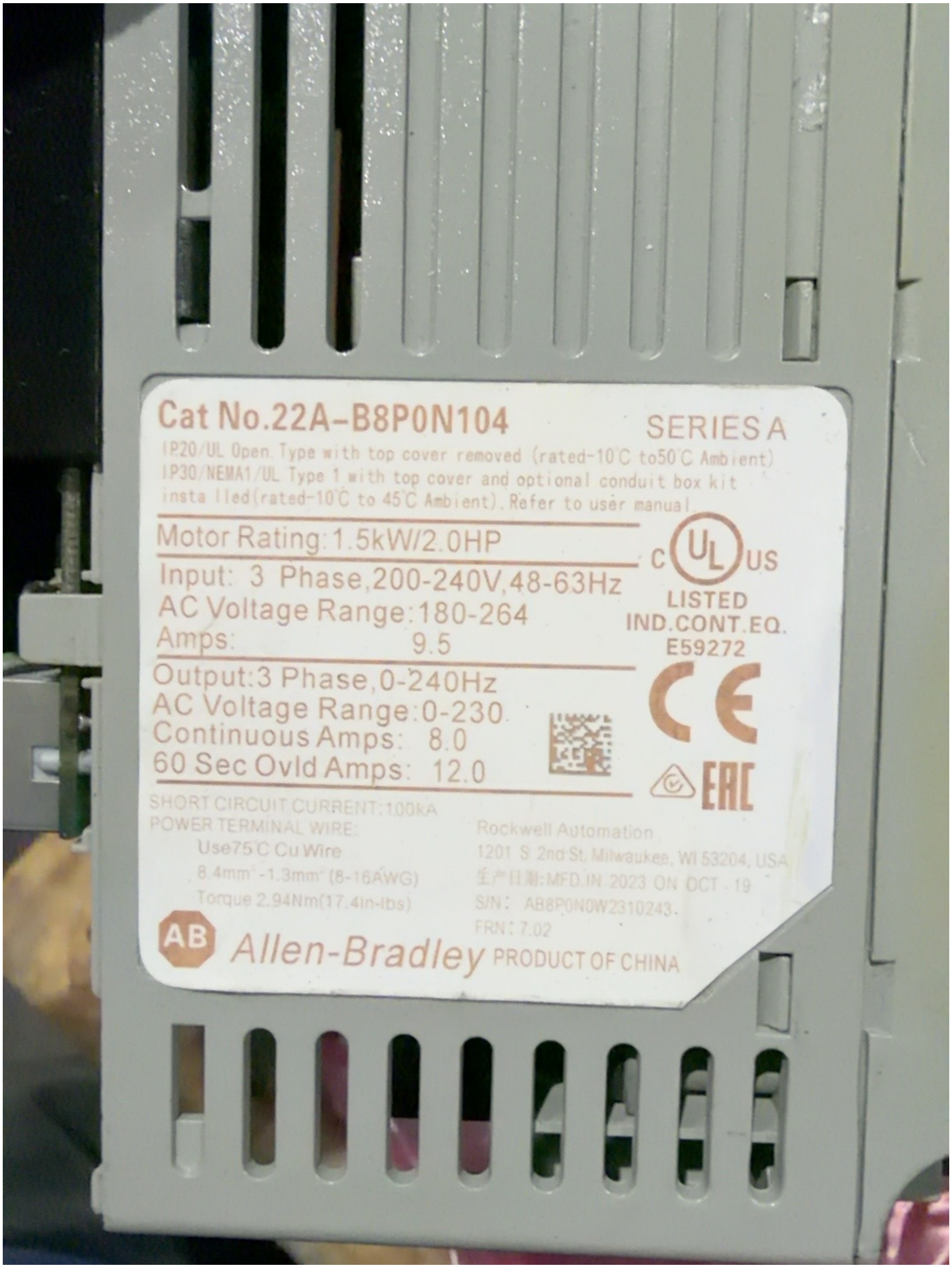
Source file: IMG_2747.jpeg | SHA-256: 0c80ffe1ede7b8434cf85d5414e552117841e43c256c2e8bcb921aafb2ce0a7c

IMG_A84788D5-2D9E-4AB2-847A-F6091267B2E2.jpeg

Allen-Bradley PowerFlex 4 nameplate; Cat. 22A-B8P0N104, Series A.



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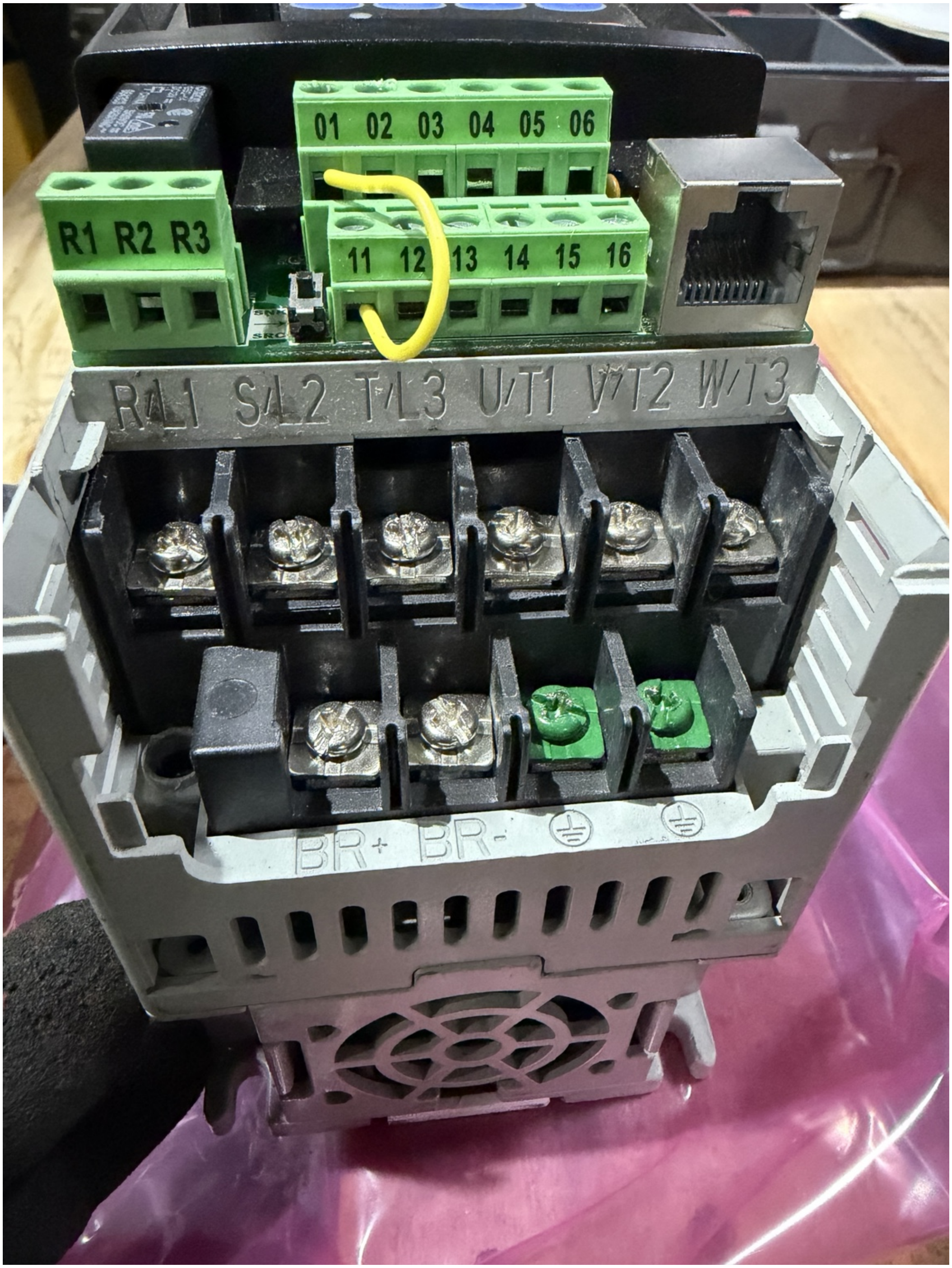
IMG_2781(1).jpeg

Allen-Bradley PowerFlex 4 front/operator interface.



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IMG_2782(1).jpeg

PowerFlex 4 fault-code and common-parameter label.

Source file: IMG_2782(1).jpeg | SHA-256: fff8983e9a61259738f3c7242e6399ca178a853a22bc96d1a43cdfc2365c6231

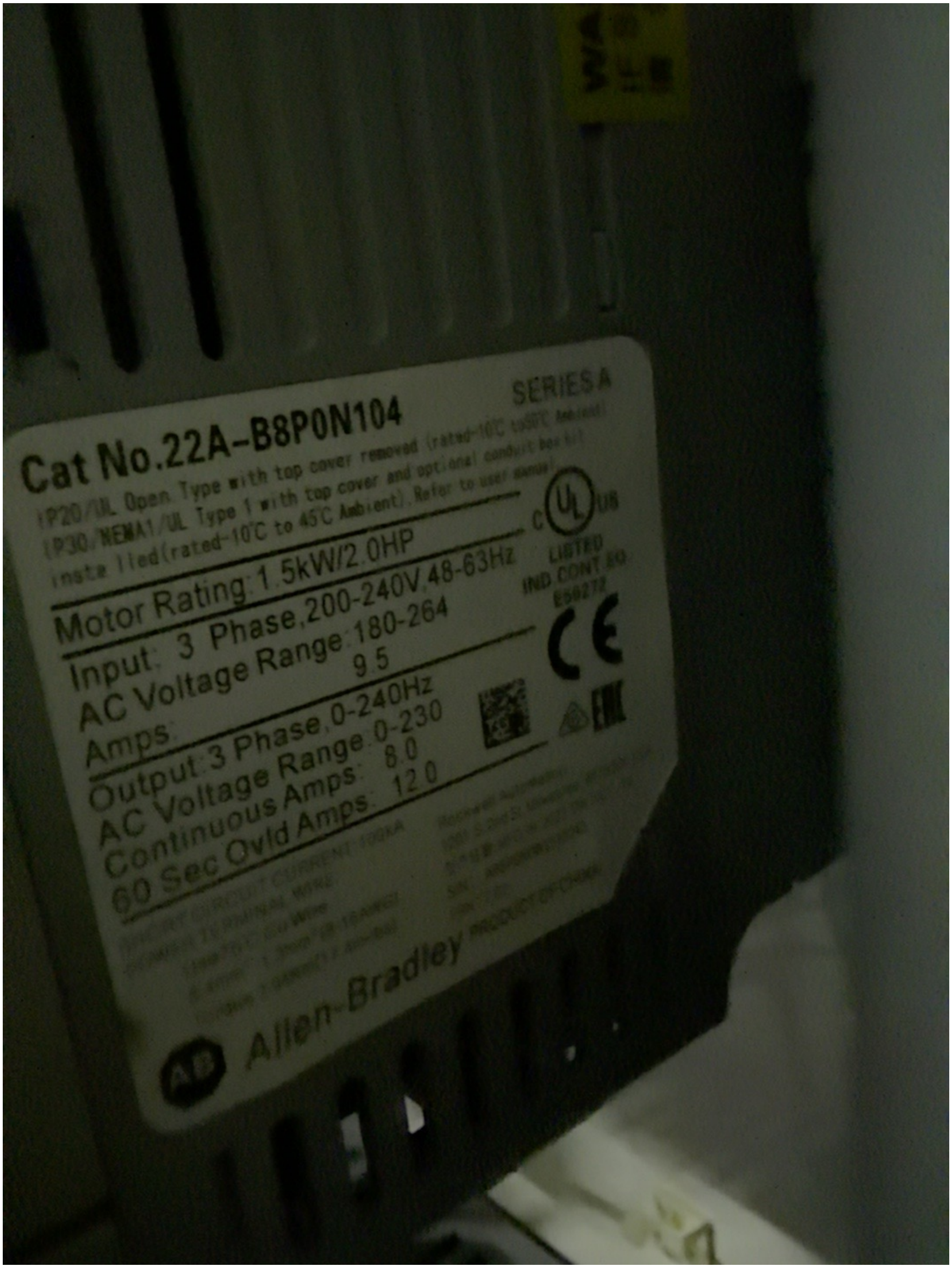
IMG_2783(1).jpeg

PowerFlex 4 power and control terminal blocks; factory jumper visible.

Source file: IMG_2783(1).jpeg | SHA-256: 0b38607f7740ce9a39daf87f55ffeb06dfcb80216a81856ad80d564c62bf8f3c

IMG_6DCB3F63-804E-40E8-B664-45F9620BD401.jpeg

PowerFlex 4 catalog/rating label alternate field view.



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IMG_2784.jpeg

PowerFlex 4 inside-cover terminal diagram and SNK/SRC definitions.

Source file: IMG_2784.jpeg | SHA-256: 9dd1bcce5589d8d0287f23d66fb8bb2b68040796bb67b295adc03f2a281e1e63

IMG_2788.jpeg

Second outfeed conveyor motor nameplate. Sterling Electric model SH0014PCA; 1 HP, 1750 RPM, 3-phase, 60 Hz, 208-230/460 V, 3.2-2.9/1.45 A, 85.5% efficiency, code K and SF 1.15. Some printed fields remain obscured by glare.

Source file: IMG_2788.jpeg | SHA-256: 556d995157fcc5ee8192e7edbd2e8d488a68268d2919454574e07a19ab0a2c9e